

LESSON 8-5 • HOMEWORK PACKET • SHEET 1 OF 2

Graphing Inequalities

Maryland Standard 6.EE.B.8 • Learning target – I can graph an inequality on a number line using an open or a closed dot.

REMEMBER

- Use an **open dot** for $<$ or $>$. Use a **closed dot** for $\leq$ or $\geq$.
- Draw the arrow toward the solutions. The dot depends on the **symbol**, not on the sign of the number.
- Test a number under your arrow. It should make the inequality true.

WORKED EXAMPLEGraph $y > -2$.**Step 1** $>$ does not include -2 , so draw an open dot at -2 .**Step 2** Solutions are greater, so the arrow points right.**Answer** Open dot at -2 , arrow right**PRACTICE***use the number lines • show your work***1.** For each graph, tell the dot (open or closed) and the arrow direction (left or right).

a. $x \geq 9$ _____

b. $n < -4$ _____

c. $p \leq 0$ _____

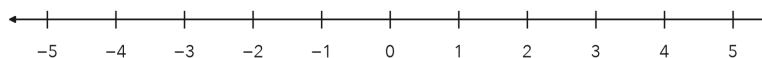
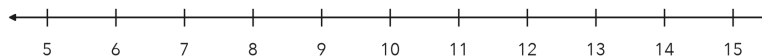
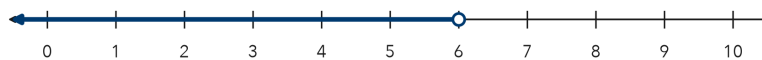
d. $k > 2.5$ _____

e. $w \geq -6$ _____

f. $r < 7$ _____

g. $b > -1.5$ _____

h. $c \leq 12$ _____

2. Graph $a \leq -1$ on the number line. Then name one solution.**One solution** _____**3. Hiking Trail.** A trail is open to hikers who are at least 10 years old. Write an inequality for a , a hiker's age. Then graph it.**Inequality** _____**4. Read the Graph.** Write the inequality shown, using m . Then describe a real situation it could show.**CHECK YOURSELF**☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____**AT HOME** Ask someone at home for a rule with a limit (bedtime, screen time, a speed limit). Write it as an inequality and graph it.

LESSON 8-5 • HOMEWORK PACKET • SHEET 2 OF 2

Testing Solutions to Inequalities

Maryland Standard 6.EE.B.8 • Learning target – I can test values against an inequality and graph the solution set.

REMEMBER

- **Substitute** the value. A **true** statement means it is a solution.
- An inequality usually has **many** solutions.
- The **boundary** is a solution for $\leq$ and $\geq$ only.

WORKED EXAMPLE

Which of 6, 8, and 11 are solutions of $x \geq 8$?**Step 1** Test 6: $6 \geq 8$ is false. Test 8: $8 \geq 8$ is true.**Step 2** Test 11: $11 \geq 8$ is true.**Answer** 8 and 11 are solutions; 6 is not.

PRACTICE

show your work • label units

1. Substitute each value. Write
- yes**
- if it is a solution or
- no**
- if it is not.

Inequality	$x = 4$	$x = 7$	$x = 10$
$x > 7$			
$x \leq 7$			
$x \geq 4$			

2. Is the value a solution? Write
- yes**
- or
- no**
- .

a. $x < 9$; $x = 9$ ____

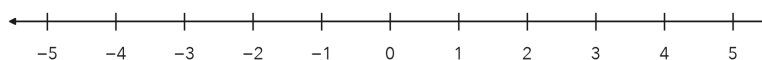
b. $m \geq -1$; $m = 0$ ____

c. $t \leq 2.5$; $t = 3$ ____

- 3.
- Bowling.**
- A bowling team wins a trophy for scoring more than 450 points. Write an inequality for the score,
- s
- . Test 450, 445, and 472.

Inequality and winning scores _____

4. Test
- -3
- ,
- 0
- , and
- 2
- in
- $y \leq -1$
- . Then graph
- $y \leq -1$
- on the number line.



Solutions from the list _____

- 5.
- Find the Error.**
- Bianca says
- $n < 12$
- has 12 solutions: 11, 10, 9, and so on down to 0. Explain what she is missing.

TEST PRACTICE

choose the best answer

6. Which value is
- not**
- a solution of
- $k \geq -2$
- ?

A -3

B -2

C 0

D 5

7. Which value is a solution of
- $n > 15$
- ?

A 12

B 14.9

C 15

D 15.5

CHECK YOURSELF

☐ I can do this on my own☐ I need a little help☐ I have a question about #_____